INCH-POUND

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8 APRIL 2009
SUPERSEDING
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16 MAY 1950

DETAIL SPECIFICATION

TERMINAL; TIE ROD, THREADED CLEVIS TYPE, AIRCRAFT

INACTIVE FOR NEW DESIGN AFTER 2 February 2001

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification establishes the requirements for one type of tie rod terminals for aircraft applications.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this standard, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

AN665	Terminal – Threaded Clevis Type Tie Rod (Inactive for new design)

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center Philadelphia (DSCP), ATTN: DSCP-NASA, 700 Robbins Avenue, Philadelphia, PA 19111-5096 or e-mail to dscpg&ispecomments@dla.mil . Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil .

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B633	Electrodeposited Coatings of Zinc on Iron and Steel
ASTM G47	Determining Susceptibility to Stress-Corrosion Cracking of 2XXX and 7XXX Aluminum Alloy Products
ASTM G69	Measurement of Corrosion Potentials of Aluminum Alloys

(Copies of these documents are available from www.astm.org or the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Terminal material shall conform to one of the following SAE steel grades: 4037, 4130, 4140, 4340, 6150, 8630 or 8740.

3.2 Design.

3.2.1 Form and Dimensions. Forms and dimensions shall be specified on the AN665.

3.2.2 Tolerances. The tapped hole, counter drilled hole and slot shall be central and parallel within the tolerances specified in AN665.

3.3 Protective Coating of Terminals. Protective coating of terminals shall be zinc plated in accordance with ASTM B633.

3.4 Tensile Strength. The tensile strength shall be no less than 115 percent of the rated strength listed on AN665 under the heading "Rated Tie Rod Strength Min (lb)".

3.5 Hardness. The hardness limits shall be within the range C26 and C31 Rockwell.

3.6 Resistance to Bend. Each prong of the fork of the terminal shall withstand a bend through the following range without cracking or indication of failure:

Terminal Size	Angle of Bend Degrees
6-4- to 1/2-20 inclusive	20
9/16 - 18 and 5/8 – 18	15

3.7 Workmanship. Shall be in accordance with high grade aircraft manufacturing practice for this type of work.

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4. VERIFICATION

4.1 Conformance inspection. Conformance inspection shall include tests 4.3.1 through 4.3.4.

4.2 Sampling. Test specimens shall be selected from the general run of production. Sufficient terminals to insure uniformity and conformance to the specified requirements shall be selected at random from each lot for test.

4.2.1 Lot. A lot shall consist of all terminals of the same type and size produced from the-same lot of steel and submitted for inspection at the same time.

4.3 Tests.

4.3.1 Examination of Product. Terminals shall be carefully inspected for conformance with the requirements of this specification with respect to workmanship, form, dimensions, threads, and marking.

4.3.2 Corrosion Resistance.- The terminals selected for inspection shall be tested for corrosion resistance in accordance with the Salt Spray tests specified in Specification ASTM G47 and ASTM G69.

4.3.3 Tensile Strength. Tensile load shall be gripped through a stay bolt screwed into the threaded end of the sample terminal, and an eyebolt or plate held by a clevis pin in the fork end of the terminal.

4.3.4 Resistance to Bending. Terminals shall be gripped by the cylindrical shank and the prongs of the fork bent through the specified angle by means of a gradually applied pressure.

4.4 Rejection and Retest.

4.4.1 Failure Under Tensile Strength or Resistance to Bend Test. If a test specimen fails to meet either of the tests, one more terminal shall be selected from the same lot. If this terminal fails to meet either or both of the required tests, the entire lot which it represents shall be rejected. The manufacturer may be allowed to reheat treat the whole lot of terminals before final tests are made. However, where failure is definitely ascribed to faulty material, no retest shall be permitted.

4.4.2 Failure Under Corrosion Resistance Tests. If any sample terminal fails to conform to the corrosion resistance requirements, the Inspector shall select from the lot, additional samples, three times as many as the number which failed. The failure of any of these additional samples shall be cause for rejection of the entire lot represented. The Inspector may, at his discretion, permit the manufacturer to strip and replate lots of tie rod terminals rejected due to failure to conform to corrosion-resistance requirements. In this case, however, the Inspector shall select samples, and witness corrosion-resistance tests of twice as many terminals from the replated lot as were originally selected as samples from the lot.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. Notes.

(This section contains Information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The terminals covered by this specification are intended for use in the construction of aircraft and aircraft accessories.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. AN part number (see AN665).
c. Packaging requirements (see 5.1).

6.3 Subject term (key word) listing.

Resistance to Bend
Zinc plated

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Navy - AS
Air Force - 99

Preparing activity:
DLA – IS

(Project 5340-2009-003)

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